EA ELITE IAL MATHEMATICS

EDEXCEL INTERNATIONAL A LEVEL

WMA12/01 January 2023 Worked Solutions

January 2023 paper rebuilt with the worked solution after each question.

10
paper questions

WMA12
Pure 2

Single-paper solutions
standalone IAL route

Dr Eslam Ahmed

Prepared for Dr Eslam Ahmed - eliteigcse.com

P2

PAST PAPER

WMA12/01 January 2023

January 2023 | 10 questions | 75 marks

10
questions

75
marks

Answers
worked solution after each question

Question 1

1.

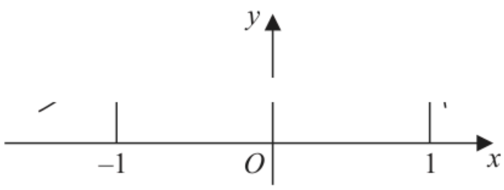

Figure 1

Figure 1 shows a sketch of part of the curve with equation $y = \mathrm{f}(x)$

The table below shows some corresponding values of x and y for this curve.

The values of y are given to 3 decimal places.

x	-1	-0.5	0	0.5	1
y	2.287	4.470	6.719	7.291	2.834

Using the trapezium rule with all the values of y in the given table,

(a) obtain an estimate for

$$\int_{-1}^{1} \mathrm{f}(x)\,\mathrm{d}x$$

giving your answer to 2 decimal places.

(3)

(b) Use your answer to part (a) to estimate

(i) $\displaystyle\int_{-1}^{1} \left(\mathrm{f}(x) - 2\right)\mathrm{d}x$

(ii) $\displaystyle\int_{1}^{3} \mathrm{f}(x-2)\,\mathrm{d}x$

(3)

(Total for Question 1 is 6 marks)

(Total for Question 1 is 6 marks)

WMA12/01 JANUARY 2023

Worked Solution - Question 1

Integration

Topic group

1. Set up the integral

Choose the correct integrand and limits, or apply the trapezium rule if the question gives tabulated values.

2. Evaluate the area or expression

Integrate or calculate the trapezium estimate carefully, then simplify the requested value.

3. State the final result

Write the answer in the form requested by the question: 6

Final answer

6

WMA12/01 JANUARY 2023

Question 2

9 marks

Integration

2. **In this question you must show all stages of your working.**

Solutions based entirely on calculator technology are not acceptable.

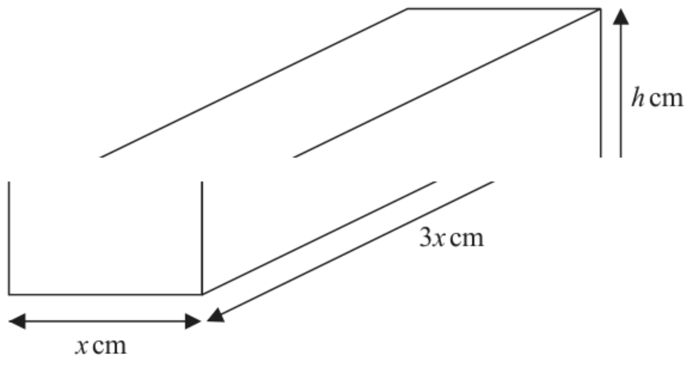

Figure 2

A brick is in the shape of a cuboid with width x cm, length $3x$ cm and height h cm, as shown in Figure 2.

The volume of the brick is $972\,\text{cm}^3$

(a) Show that the surface area of the brick, $S\,\text{cm}^2$, is given by

$$S = 6x^2 + \frac{2592}{x}$$

(3)

(b) Find $\dfrac{dS}{dx}$

(1)

(c) Hence find the value of x for which S is stationary.

(2)

(d) Find $\dfrac{d^2S}{dx^2}$ and hence show that the value of x found in part (c) gives the minimum value of S.

(2)

(e) Hence find the minimum surface area of the brick.

(1)

(Total for Question 2 is 9 marks)

(Total for Question 2 is 9 marks)

WMA12/01 JANUARY 2023

Worked Solution - Question 2

Integration

Topic group

1. Set up the integral

Choose the correct integrand and limits, or apply the trapezium rule if the question gives tabulated values.

2. Evaluate the area or expression

Integrate or calculate the trapezium estimate carefully, then simplify the requested value.

3. State the final result

Write the answer in the form requested by the question: + =

Final answer

+ =

Question 3

7 marks

Binomial Expansion

3. $$f(x) = \left(2 + \frac{kx}{8}\right)^7 \quad \text{where } k \text{ is a non-zero constant}$$

(a) Find the first 4 terms, in ascending powers of x, of the binomial expansion of $f(x)$. Give each term in simplest form.

(4)

Given that, in the binomial expansion of $f(x)$, the coefficients of x, x^2 and x^3 are the first 3 terms of an arithmetic progression,

(b) find, using algebra, the possible values of k.

(*Solutions relying entirely on calculator technology are not acceptable.*)

(3)

(Total for Question 3 is 7 marks)

(Total for Question 3 is 7 marks)

WMA12/01 JANUARY 2023

Worked Solution - Question 3

Binomial Expansion

Topic group

1. Choose the required binomial terms

Write the expansion term-by-term and keep only the powers needed for the coefficient or approximation.

2. Compare coefficients

Collect like powers, compare with the given expression, and solve for the required constant.

3. State the final result

Write the answer in the form requested by the question: - = 16 16, 5 k =

Final answer

- = 16 16, 5 k =

WMA12/01 JANUARY 2023

Question 4

6 marks

Laws of Logarithms

4. (i) Using the laws of logarithms, solve

$$\log_3(4x) + 2 = \log_3(5x + 7)$$

(3)

(ii) Given that

$$\sum_{r=1}^{2} \log_a\left(y^r\right) = \sum_{r=1}^{2} \left(\log_a y\right)^r \qquad y > 1,\ a > 1,\ y \neq a$$

find y in terms of a, giving your answer in simplest form.

(3)

(Total for Question 4 is 6 marks)

(Total for Question 4 is 6 marks)

WMA12/01 JANUARY 2023

Worked Solution - Question 4

Laws of Logarithms

Topic group

1. Combine the logarithms

Use the power, product, and quotient laws to rewrite the equation as a single logarithmic or exponential statement.

2. Solve with domain checks

Solve the resulting equation and reject any value that does not satisfy the original logarithms.

3. State the final result

Write the answer in the form requested by the question: The required value or expression is obtained by the final simplification.

Final answer

The required value or expression is obtained by the final simplification.

WMA12/01 JANUARY 2023

Question 5

9 marks

Polynomials

5. $$f(x) = x^3 + (p + 3)x^2 - x + q$$

where p and q are constants and $p > 0$

Given that $(x - 3)$ is a factor of $f(x)$

(a) show that

$$9p + q = -51$$

(2)

Given also that when $f(x)$ is divided by $(x + p)$ the remainder is 9

(b) show that

$$3p^2 + p + q - 9 = 0$$

(2)

(c) Hence find the value of p and the value of q.

(3)

(d) Hence find a quadratic expression $g(x)$ such that

$$f(x) = (x - 3)g(x)$$

(2)

(Total for Question 5 is 9 marks)

(Total for Question 5 is 9 marks)

WMA12/01 JANUARY 2023

Worked Solution - Question 5

Polynomials

Topic group

1. Use the polynomial condition

Apply the factor theorem, remainder theorem, or substitution stated in the question to form the required equation.

2. Finish the algebra

Solve the resulting equation and substitute back where needed, then write the requested factor, coefficient, or value.

3. State the final result

Write the answer in the form requested by the question: - + + ( ) 2 g 12 35 x x x = + +

Final answer

- + + ( ) 2 g 12 35 x x x = + +

WMA12/01 JANUARY 2023

Question 6

10 marks

Circles

6. The circle C has equation

$$x^2 + y^2 + 8x - 4y = 0$$

(a) Find

(i) the coordinates of the centre of C,

(ii) the exact radius of C.

(3)

The point P lies on C.

Given that the tangent to C at P has equation $x + 2y + 10 = 0$

(b) find the coordinates of P

(4)

(c) Find the equation of the normal to C at P, giving your answer in the form $y = mx + c$ where m and c are integers to be found.

(3)

(Total for Question 6 is 10 marks)

(Total for Question 6 is 10 marks)

WMA12/01 JANUARY 2023

Worked Solution - Question 6

Circles

Topic group

1. Identify the circle information

Use the centre-radius form, distance formula, gradient condition, or tangent property required by the diagram.

2. Complete the geometry

Substitute the coordinates carefully and simplify to the requested equation, length, point, or proof.

3. State the final result

Write the answer in the form requested by the question: + or ( ) 2 2 6 y x + = + 2 10 y x = +

Final answer

+ or ( ) 2 2 6 y x + = + 2 10 y x = +

Question 7

7 marks

Geometric Sequences

7. A geometric sequence has first term a and common ratio r, where $r > 0$

Given that

- the 3rd term is 20
- the 5th term is 12.8

(a) show that $r = 0.8$

(1)

(b) Hence find the value of a.

(2)

Given that the sum of the first n terms of this sequence is greater than 156

(c) find the smallest possible value of n.

(*Solutions based entirely on graphical or numerical methods are not acceptable.*)

(4)

(Total for Question 7 is 7 marks)

(Total for Question 7 is 7 marks)

WMA12/01 JANUARY 2023

Worked Solution - Question 7

Geometric Sequences

Topic group

1. Use the geometric formula

Identify the first term, common ratio, and whether the question needs a term, finite sum, or sum to infinity.

2. Substitute and solve

Apply the correct geometric formula and simplify, checking any validity condition on the ratio.

3. State the final result

Write the answer in the form requested by the question: $= 31$

Final answer

$= 31$

Question 8

9 marks

Trigonometric Equations

8. **In this question you must show all stages of your working.**

Solutions based entirely on calculator technology are not acceptable.

(i) Solve, for $-\frac{\pi}{2} < x < \pi$, the equation

$$5\sin(3x + 0.1) + 2 = 0$$

giving your answers, **in radians**, to 2 decimal places.

(4)

(ii) Solve, for $0 < \theta < 360°$, the equation

$$2\tan\theta\sin\theta = 5 + \cos\theta$$

giving your answers, **in degrees**, to one decimal place.

(5)

(Total for Question 8 is 9 marks)

(Total for Question 8 is 9 marks)

WMA12/01 JANUARY 2023

Worked Solution - Question 8

Trigonometric Equations

Topic group

1. Rearrange the trigonometric equation

Use the identity or ratio needed to reduce the equation to one trigonometric function in the required interval.

2. List the valid solutions

Find all angles in the given range and discard any extra values outside the interval.

3. State the final result

Write the answer in the form requested by the question: The required value or expression is obtained by the final simplification.

Final answer

The required value or expression is obtained by the final simplification.

WMA12/01 JANUARY 2023

Question 9

8 marks

Integration

9. **In this question you must show all stages of your working.**

Solutions based entirely on calculator technology are not acceptable.

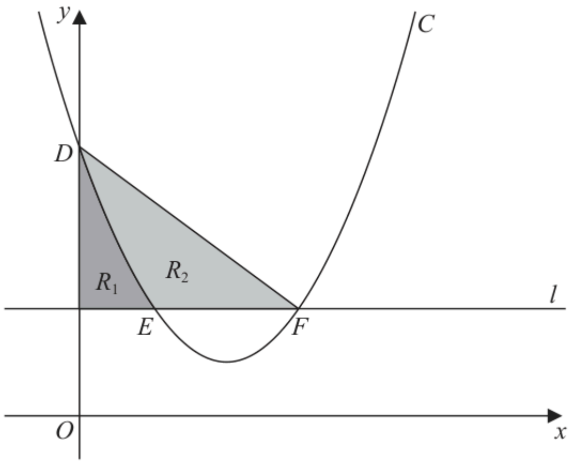

Figure 3

Figure 3 shows

- the curve C with equation $y = x^2 - 4x + 5$
- the line l with equation $y = 2$

The curve C intersects the y-axis at the point D.

(a) Write down the coordinates of D.

(1)

The curve C intersects the line l at the points E and F, as shown in Figure 3.

(b) Find the x coordinate of E and the x coordinate of F.

(2)

Shown shaded in Figure 3 is

- the region R_1 which is bounded by C, l and the y-axis
- the region R_2 which is bounded by C and the line segments EF and DF

Given that $\dfrac{\text{area of } R_1}{\text{area of } R_2} = k$, where k is a constant,

(c) use algebraic integration to find the exact value of k, giving your answer as a simplified fraction.

(5)

(Total for Question 9 is 8 marks)

(Total for Question 9 is 8 marks)

WMA12/01 JANUARY 2023

Worked Solution - Question 9

Integration

Topic group

1. Set up the integral

Choose the correct integrand and limits, or apply the trapezium rule if the question gives tabulated values.

2. Evaluate the area or expression

Integrate or calculate the trapezium estimate carefully, then simplify the requested value.

3. State the final result

Write the answer in the form requested by the question: The required value or expression is obtained by the final simplification.

Final answer

The required value or expression is obtained by the final simplification.

Question 10

4 marks

Proof

10. A student was asked to prove by exhaustion that

if n is an integer then $2n^2 + n + 1$ is **not** divisible by 3

The start of the student's proof is shown in the box below.

Consider the case when $n = 3k$ $$2n^2 + n + 1 = 18k^2 + 3k + 1 = 3(6k^2 + k) + 1$$ which is not divisible by 3

Complete this proof.

(4)

(Total for Question 10 is 4 marks)

(Total for Question 10 is 4 marks)

WMA12/01 JANUARY 2023

Worked Solution - Question 10

Proof

Topic group

1. Set up the proof

Translate the statement into algebra, or choose a clear counterexample if the statement is false. Keep each implication justified.

2. Complete the argument

Simplify the algebra or evaluate the counterexample, then finish with a clear conclusion.

3. State the final result

Write the answer in the form requested by the question: - + which is not divisible by 3 Hence 2 2 1 n n + + is not divisible by 3

Final answer

- + which is not divisible by 3 Hence 2 2 1 n n + + is not divisible by 3